



Aakash

Medical | IIT-JEE | Foundations

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AIM - 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

CST-I

Time : 180 Mins.

Answers

1. (3)	37. (4)	73. (3)	109. (4)	145. (4)
2. (2)	38. (3)	74. (1)	110. (1)	146. (3)
3. (3)	39. (4)	75. (1)	111. (2)	147. (3)
4. (3)	40. (4)	76. (2)	112. (3)	148. (1)
5. (3)	41. (4)	77. (4)	113. (2)	149. (3)
6. (2)	42. (3)	78. (4)	114. (2)	150. (1)
7. (2)	43. (2)	79. (3)	115. (1)	151. (4)
8. (1)	44. (4)	80. (1)	116. (3)	152. (4)
9. (4)	45. (4)	81. (1)	117. (3)	153. (4)
10. (3)	46. (2)	82. (2)	118. (3)	154. (2)
11. (1)	47. (1)	83. (2)	119. (3)	155. (2)
12. (2)	48. (1)	84. (3)	120. (1)	156. (3)
13. (4)	49. (3)	85. (2)	121. (2)	157. (1)
14. (4)	50. (3)	86. (2)	122. (3)	158. (1)
15. (4)	51. (2)	87. (4)	123. (4)	159. (4)
16. (3)	52. (3)	88. (1)	124. (3)	160. (3)
17. (1)	53. (4)	89. (2)	125. (2)	161. (2)
18. (3)	54. (1)	90. (4)	126. (3)	162. (3)
19. (2)	55. (1)	91. (1)	127. (3)	163. (3)
20. (2)	56. (3)	92. (1)	128. (2)	164. (2)
21. (2)	57. (4)	93. (3)	129. (3)	165. (4)
22. (2)	58. (4)	94. (3)	130. (1)	166. (3)
23. (4)	59. (4)	95. (3)	131. (3)	167. (4)
24. (3)	60. (2)	96. (2)	132. (2)	168. (3)
25. (3)	61. (2)	97. (3)	133. (3)	169. (2)
26. (4)	62. (1)	98. (4)	134. (3)	170. (1)
27. (2)	63. (2)	99. (4)	135. (2)	171. (4)
28. (2)	64. (4)	100. (4)	136. (4)	172. (4)
29. (1)	65. (1)	101. (3)	137. (3)	173. (3)
30. (3)	66. (3)	102. (3)	138. (3)	174. (2)
31. (3)	67. (3)	103. (4)	139. (4)	175. (3)
32. (3)	68. (4)	104. (3)	140. (3)	176. (3)
33. (4)	69. (3)	105. (4)	141. (3)	177. (2)
34. (1)	70. (1)	106. (1)	142. (3)	178. (3)
35. (3)	71. (3)	107. (4)	143. (2)	179. (3)
36. (2)	72. (3)	108. (4)	144. (3)	180. (4)

Hints and Solutions

PHYSICS

(1) Answer : (3)

Solution:

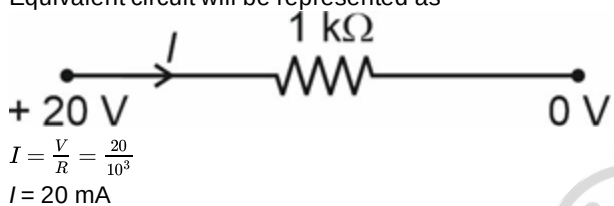
A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

Therefore, NOR gate

(2) Answer : (2)

Solution:

Equivalent circuit will be represented as



(3) Answer : (3)

Solution:

$$n = \frac{P}{\frac{hc}{\lambda}}$$

$$n = \frac{1.6 \times 10^{-7}}{\frac{12400}{1240} \times 1.6 \times 10^{-19}}$$

$$n = \frac{1.6 \times 10^{-7} \times 1240 \times 10^{19}}{12400 \times 1.6}$$

$$n = 10^{11} \text{ photons/second}$$

(4) Answer : (3)

Solution:

Binding energy of an isolated neutron is zero.

$$Q = (\text{B.E.})_P - (\text{B.E.})_R$$

$$= c - a - b$$

(5) Answer : (3)

Solution:

$$v = f\lambda$$

$$\lambda = \frac{330}{1320} = \frac{1}{4} \text{ m}$$

$$\text{First resonance } \frac{\lambda}{4} = \frac{1}{16} \text{ m}$$

$$n^{\text{th}} \text{ resonance } = \frac{(2n-1)\lambda}{4} = \frac{120}{100}$$

$$(2n-1) = \frac{120 \times 16}{100}$$

$$2n-1 = 19.2$$

$$n = 10$$

(6) Answer : (2)

Solution:

Power of one part:

$$\text{Power } P = \frac{1}{f} = (\mu - 1) \left[\frac{1}{\infty} - \left(-\frac{1}{R} \right) \right] = \frac{\mu - 1}{R}$$

(7) Answer : (2)

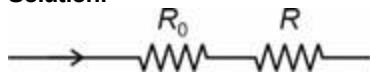
Solution:

Statement A is wrong because centre of gravity of a body is the point where total gravitation torque on body is zero.

Statement B is wrong because for symmetric bodies, the centre of mass always coincides with their geometric centre and not with its centre of gravity in non-uniform gravitational field.

(8) Answer : (1)

Solution:



$$V = IR_0$$

$$(R_0 + R)I = 3V$$

$$IR = 2V$$

$$IR = 2R_0I$$

$$R = 2R_0$$

(9) Answer : (4)

Solution:

$$g = \frac{GM}{R^2} = G \left(\rho \frac{4}{3} \pi R^3 \right) \times \frac{1}{R^2} = \frac{4\pi\rho RG}{3}$$

$$\Rightarrow g \propto \rho R$$

$$\therefore \frac{g_1}{g_2} = \frac{\rho_1}{\rho_2} \times \frac{R_1}{R_2} = \frac{3}{8} \times \frac{2}{3} = \frac{1}{4}$$

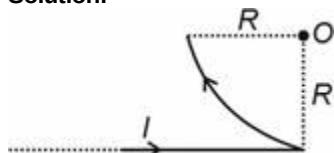
(10) Answer : (3)

Solution:

EMF will be induced in the wire if length, velocity and magnetic field are mutually perpendicular.

(11) Answer : (1)

Solution:



$$B = B_1 + B_2$$

$$B_1 = \frac{\mu_0 i}{4\pi R} \hat{k}$$

$$B_2 = \frac{\mu_0 i}{8R} (-\hat{k})$$

$$B = \left(\frac{\mu_0 i}{4\pi R} - \frac{\mu_0 i}{8R} \right) \hat{k}$$

(12) Answer : (2)

Solution:

$$M_{net} = -\frac{ml}{2} \hat{i} + \frac{ml}{2} \hat{i} + \frac{\sqrt{3}ml}{2} \hat{j}$$

$$= \frac{\sqrt{3}ml}{2} \hat{j}$$

(13) Answer : (4)

Solution:

Voltage in AC circuit may lead, lag or in phase depends on capacitive and inductive reactance of the circuit.

(14) Answer : (4)

Solution:

A body can have constant speed but variable velocity while moving on any path other than straight line.

If a body is moving with constant acceleration but variable velocity then its path may or may not be a straight line.

(15) Answer : (4)

Solution:

The specific volume is minimum at 4°C .

Liquids only have volumetric expansion.

For heat transfer, temperature difference must be there.

(16) Answer : (3)

Solution:

Since, $\Delta V = 0$

$\Rightarrow$ Process will be isochoric.

(17) Answer : (1)

Solution:

For an adiabatic process, $\Delta Q = 0$

$$\Rightarrow \Delta U + \Delta W = 0$$

$$\Rightarrow \Delta U = -\Delta W$$

For an isothermal process, $\Delta T = 0$

$$\Rightarrow \Delta U = 0$$

(18) Answer : (3)

Solution:

$$\Delta P = \frac{4T}{r} \Rightarrow \Delta P \propto \frac{1}{r}$$

as $r \downarrow \Delta P \uparrow$

(19) Answer : (2)

Solution:

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow v_{\text{rms}} \propto \sqrt{T}$$

$$\Rightarrow \frac{v}{v'} = \sqrt{\frac{400}{500}}$$

$$\Rightarrow \frac{v}{v'} = \frac{2}{\sqrt{5}}$$

$$\Rightarrow v' = \sqrt{\frac{5}{4}} v$$

(20) Answer : (2)

Solution:

Young's modulus is defined for solids only.

$Y_{\text{steel}} > Y_{\text{rubber}}$

$\therefore$ Steel is more elastic than rubber.

(21) Answer : (2)

Solution:

KE = PE

$$\frac{1}{2} m \omega^2 (A^2 - y^2) = \frac{1}{2} m \omega^2 y^2$$

$$A^2 = 2y^2$$

$$y = \frac{A}{\sqrt{2}} = 0.7074 = 70.7\% \text{ of } A$$

(22) Answer : (2)

Solution:

Standing wave $y = 2A \sin kx \sin \omega t$

Given wave $y = 20 \sin(4\pi x) \sin(20\pi t)$

$$2A = 20, A = 10$$

$$k = 4\pi = \frac{2\pi}{\lambda}, \lambda = 0.5$$

$$\text{Velocity of wave } v = \frac{\omega}{k} = \frac{20\pi}{4\pi} = 5$$

Separation between nodes and antinodes

$$= \frac{\lambda}{4} = 0.125$$

(23) Answer : (4)

Solution:

$$\text{Energy of photon } U = hf = \frac{hc}{\lambda}$$

$$\text{Wavelength of photon } \lambda = \frac{h}{p}$$

$$\therefore \boxed{U = pc}$$

(24) Answer : (3)

Solution:

$$I_{\text{resultant}} = 4I \cos^2 \left(\frac{\phi}{2} \right)$$

$$\text{Fev } \phi = \frac{\pi}{3}$$

$$I_{\text{resultant}} = 4I \cos^2 \left(\frac{\pi}{6} \right)$$

$$= 4I \left(\frac{\sqrt{3}}{2} \right)^2 = 3I$$

(25) Answer : (3)

Solution:

$$[r] = [v] [s + t]$$

$$= [LT^{-1}] [T]$$

$$[r] = [L]$$

(26) Answer : (4)

Solution:

$$v_{avg} = \frac{\Delta s}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$

$$= \frac{[2(4) + 4 \times 2 - 6] - [2(0)^2 + 4(0) - 6]}{2}$$

$$= 8 \text{ m/s}$$

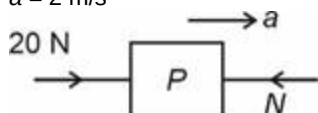
(27) Answer : (2)

Solution:

$$F_{net} = ma$$

$$20 = 10a$$

$$a = 2 \text{ m/s}^2$$



$$20 - N = 5 \times 2$$

$$N = 20 - 10$$

$$= 10 \text{ N}$$

(28) Answer : (2)

Solution:

Change in velocity = Area under $a-t$ curve

$$v_f - v_i = \frac{1}{2} \times 4 \times 4$$

$$v_f = 8 + 4$$

$$= 12 \text{ m/s}$$

(29) Answer : (1)

Solution:

$$E_i = E_f$$

$$0 = 5gh - 15gh + \frac{1}{2}(5 + 15)v^2$$

$$10gh = \frac{1}{2} \times 20v^2$$

$$v = 10 \text{ m/s}$$

(30) Answer : (3)

Solution:

W.D. by conservative force ($W_{cons} = U_i - U_f$) and is dependent on end points only.

(31) Answer : (3)

Solution:

$$F = \frac{kq_1q_2}{r^2} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(10^{-10})^2} \approx 23 \times 10^{-9} \text{ N}$$

(32) Answer : (3)

Solution:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{3 \times 6}{3 + 6} = 2 \mu\text{F}$$

After a long time, charge stored in the capacitor is

$$Q = CV \Rightarrow Q = 2 \times 6 = 12 \mu\text{C}$$

Work done by the battery

$$W_b = q\Delta V$$

$$W_b = 12 \times 6 = 72 \mu\text{J}$$

(33) Answer : (4)

Solution:

$$I = neAv_d \Rightarrow I \propto v_d$$

If v_d increase then I increases

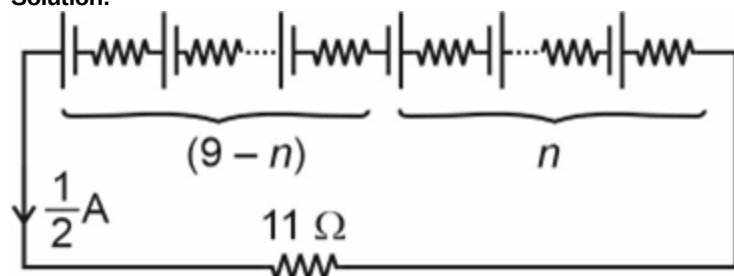
$$v_d = \frac{eE}{m} \tau$$

$$\text{and } \tau \propto \frac{1}{T}$$

Therefore, for constant electric field $v_d \propto \frac{1}{T}$

(34) Answer : (1)

Solution:



$$i = \frac{\text{net emf}}{\text{Total resistance}}$$

$$i = \frac{(9-n)\varepsilon - n\varepsilon}{R + 9r}$$

$$\frac{1}{2} = \frac{(9-2n) \times 2}{20} \Rightarrow 10 = (9-2n)2 \rightarrow n = 2$$

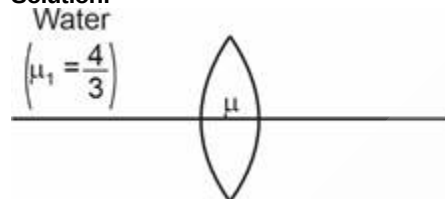
(35) Answer : (3)

Solution:

Large focal length of objective lens enhances the magnifying power of the telescope. The objective also has larger aperture so that it can collect more light.

(36) Answer : (2)

Solution:



Using lens maker's formula,

$$\frac{1}{f} = \left(\frac{\mu}{\mu_0} - 1 \right) \left[\frac{1}{R} - \left(-\frac{1}{R} \right) \right]$$

$$= \left(\frac{\frac{3}{2}}{\frac{4}{3}} - 1 \right) \frac{2}{R}$$

$$= \frac{1}{8} \times \frac{2}{R} = \frac{1}{4R}$$

$$\therefore f = 4R = 120 \text{ cm}$$

(37) Answer : (4)

Solution:

Inside the shell,

$$V_P = -\frac{GM}{R}$$

everywhere independent of the location of point P .

(38) Answer : (3)

Solution:

$$\text{Moment of inertia of disc about } P = \frac{MR^2}{2} + M(1)^2 = \frac{10(2)^2}{2} + 10(1)^2$$

$$\Rightarrow I_P = 30 \text{ kg m}^2$$

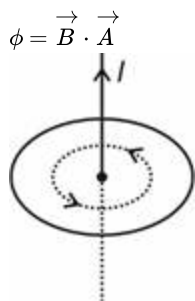
$$\text{Torque about } P = I_P \alpha$$

$$\Rightarrow 1 \times 18 = 30\alpha$$

$$\Rightarrow \alpha = \frac{3}{5} \text{ rad/s}^2$$

(39) Answer : (4)

Solution:



$$\phi = BA \cos 90^\circ$$

$$= \text{Zero}$$

(40) Answer : (4)

Solution:

$$I = \frac{V}{Z}$$

$$Z = \frac{V}{I}$$

$$= \frac{100}{20}$$

$$= 5 \Omega$$

$$\cos \phi = \frac{R}{Z}$$

$$\cos 60^\circ = \frac{R}{5} \Rightarrow R = \frac{5}{2} \Omega$$

(41) Answer : (4)

Solution:

In third orbit circumference is equal to 3λ

$$3 \times 10^{-8} \text{ m}$$

(42) Answer : (3)

Solution:

$$1 \text{ ppm} = 1 \text{ in } 10^6 \text{ atoms}$$

$$\therefore \text{Number of electrons } (n_e) = \frac{4 \times 10^{26}}{10^6}$$

$$= 4 \times 10^{20} \text{ m}^{-3}$$

$$\text{Number of holes } (n_h) = \frac{n_i^2}{n_e}$$

$$= \frac{1.4 \times 10^{16} \times 1.4 \times 10^{16}}{4 \times 10^{20}}$$

$$= 49 \times 10^{-2} \times 10^{32} \times 10^{-20}$$

$$= 49 \times 10^{10} \text{ m}^{-3}$$

(43) Answer : (2)

Solution:

$$\text{Excess force} = 2 \times T \times 2\pi r$$

$$= 2 \times \frac{7}{100} \times 2 \times \frac{22}{7} \times 4.5 \times 10^{-2}$$

$$= 39.6 \text{ mN}$$

(44) Answer : (4)

Solution:

$$\text{Power of star} = \sigma T^4 \times 4\pi R^2 = Q$$

$$\Rightarrow T = \left(\frac{Q}{4\pi R^2 \sigma} \right)^{\frac{1}{4}}$$

(45) Answer : (4)

Solution:

$$\text{Width of central maxima} = \frac{2\lambda D}{a}$$

$$= \frac{2 \times 5 \times 10^{-7} \times 2}{2 \times 10^{-3}}$$

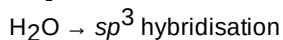
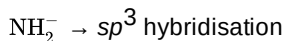
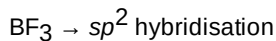
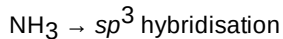
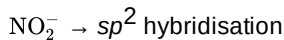
$$= 10^{-3} \text{ m}$$

$$= 1 \text{ mm}$$

CHEMISTRY

(46) Answer : (2)

Solution:



(47) Answer : (1)

Solution:

Smaller the size of cations higher is its polarising power and higher is covalent character.

(48) Answer : (1)

Solution:

$$\Delta T_f = K_f \times m$$

$$= \frac{1.86 \times 34.2 \times 1000}{342 \times 1000}$$

$$= 0.186$$

(49) Answer : (3)

Solution:

According to Le Chatelier's principle solubility of gases in liquids decreases with rise in temperature.

(50) Answer : (3)

Solution:

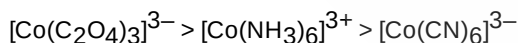
N_2O is not an odd electron molecule.

(51) Answer : (2)

Solution:

Order of field strength of ligands : $\text{CN}^- > \text{NH}_3 > \text{C}_2\text{O}_4^{2-}$

field strength of ligand increases, Crystal field splitting, Δ_0 also increases, wavelength decreases



(52) Answer : (3)

Solution:

Stability of carbocations decides the increasing order of reactivity towards SN_1 reaction.

(53) Answer : (4)

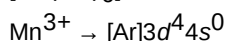
Solution:

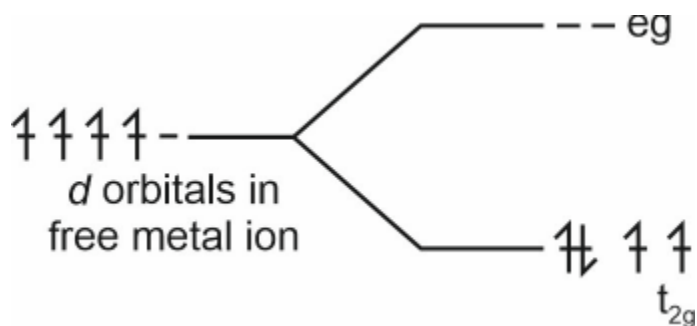
Amine	pK_b value
Benzenamine	9.38
Phenylmethanamine	4.70
N-Methylaniline	9.30
N,N-Dimethylaniline	8.92

(54) Answer : (1)

Solution:

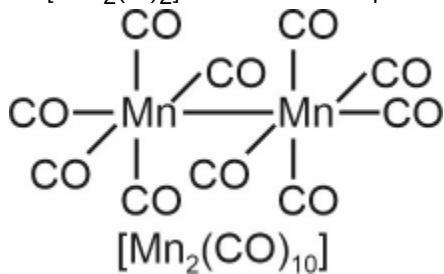
$[\text{Mn}(\text{CN})_6]^{3-}$ has two unpaired electrons and d^2sp^3 hybridisation.





CN^- is strong field ligand which have $\Delta_0 > P$

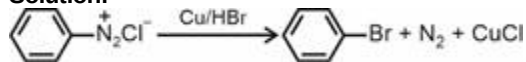
$\text{cis}[\text{CrCl}_2(\text{ox})_2]^{3-}$ does not have plane of symmetry so it is optically active.



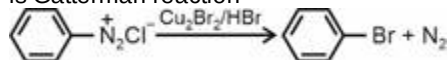
It has one Mn – Mn bond and zero bridged CO group.

(55) Answer : (1)

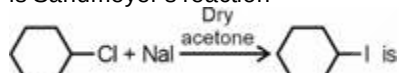
Solution:



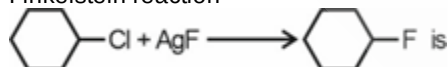
is Gatterman reaction



is Sandmeyer's reaction



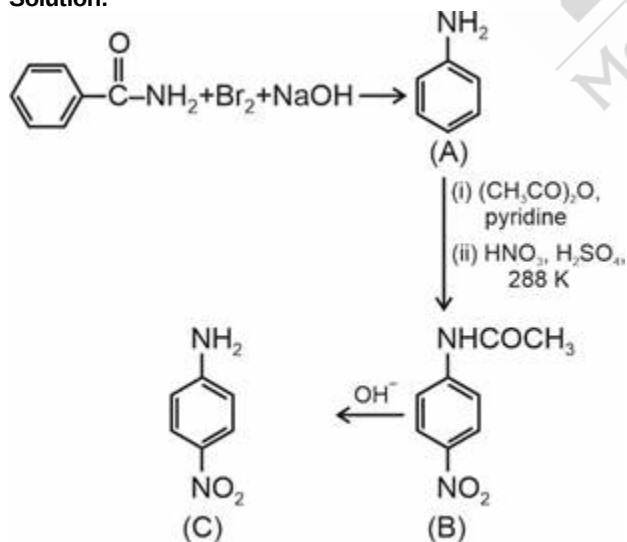
Finkelstein reaction



Swarts reaction

(56) Answer : (3)

Solution:



(57) Answer : (4)

Solution:

Kernite $\text{Na}_2\text{B}_4\text{O}_7 \times 4\text{H}_2\text{O}$

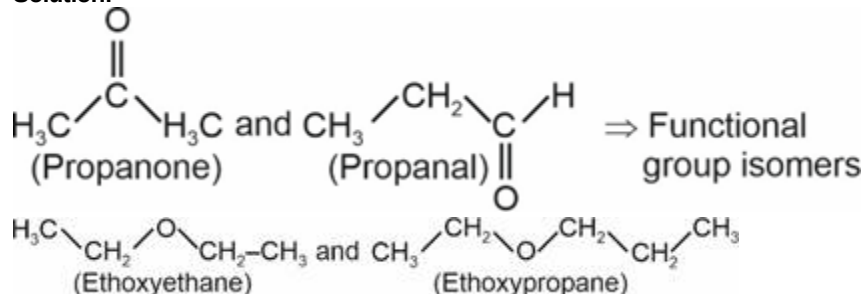
 Borax $\text{Na}_2\text{B}_4\text{O}_7 \times 10\text{H}_2\text{O}$

 Bauxite $\text{Al}_2\text{O}_3 \times 2\text{H}_2\text{O}$

 Cryolite Na_3AlF_6

(58) Answer : (4)

Solution:

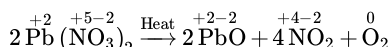
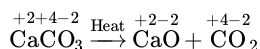


are not metamers

(59) Answer : (4)

Solution:

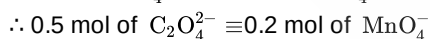
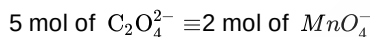
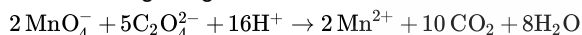
The redox reaction in which same element gets oxidised as well as reduced simultaneously.


 $\Rightarrow$ Not a disproportionation

 $\Rightarrow$ Not a redox reaction

(60) Answer : (2)

Solution:

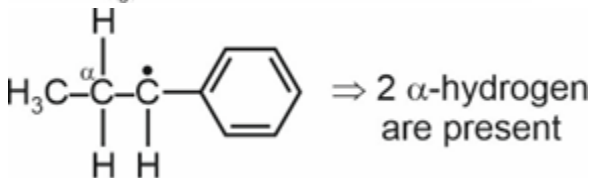
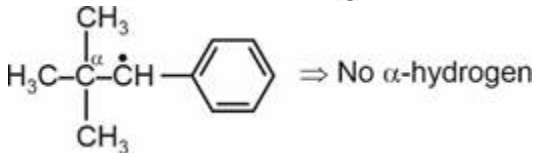
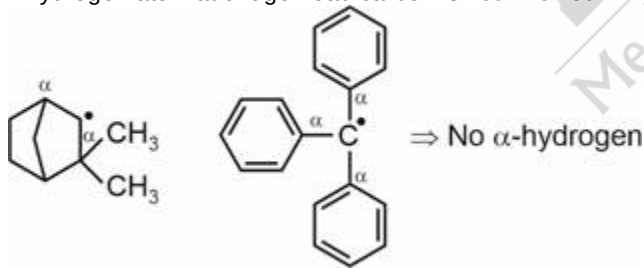
On balancing the given redox reaction:



(61) Answer : (2)

Solution:

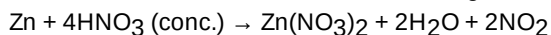
- If hydrogen atom is present at α carbon to the free radical then hyper conjugation is possible.
- Hydrogen atom at bridge head carbon is not involved in hyperconjugation.



(62) Answer : (1)

Solution:

Zinc reacts with concentrated nitric acid to give NO_2



(63) Answer : (2)

Solution:

$\text{COCl}_2 \rightarrow$ Phosgene

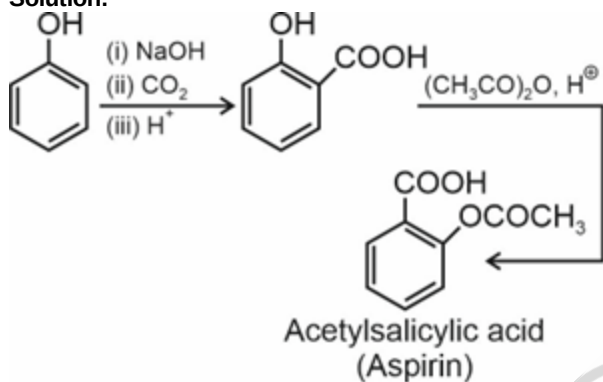
$\text{CCl}_3\text{NO}_2 \rightarrow$ Tear gas

$\text{ClCH}_2\text{CH}_2\text{SCH}_2\text{CH}_2\text{Cl} \rightarrow$ Mustard gas

$\text{CHCl}_3 \rightarrow$ Chloroform

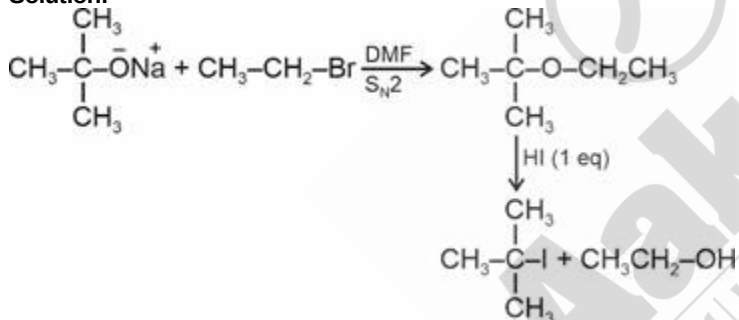
(64) Answer : (4)

Solution:



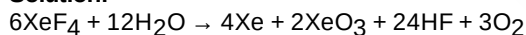
(65) Answer : (1)

Solution:



(66) Answer : (3)

Solution:



(67) Answer : (3)

Solution:

Compound **pK_a**

o-Nitrophenol – 7.2

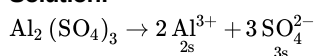
p-Nitrophenol – 7.1

m-Cresol – 10.1

p-Cresol – 10.2

(68) Answer : (4)

Solution:



$$K_{sp} = (\text{Al}^{3+})^2 (\text{SO}_4^{2-})^3$$

$$K_{sp} = (2s)^2 (3s)^3$$

$$K_{sp} = 4s^2 \times 27s^3$$

$$K_{sp} = 108s^5$$

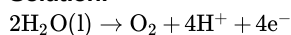
(69) Answer : (3)

Solution:

Metallic character and reactivity increases down the group and decreases on moving from left to right across the period.

(70) Answer : (1)

Solution:



$$n \times n_F = \frac{q}{F}$$

$$\frac{16}{32} \times 4 = \frac{q}{F}$$

$$2F = q$$

(71) Answer : (3)

Solution:

$$K_W = K_a \times K_b$$

$$10^{-14} = 10^{-6} \times K_b$$

$$K_b = \frac{10^{-14}}{10^{-6}}$$

$$K_b = 10^{-8}$$

(72) Answer : (3)

Solution:

$$\Lambda_m = \frac{K}{C} \times 1000$$

(If K is in $S\,cm^{-1}$)

$$\Lambda_m = \frac{1.6 \times 10^{-3}}{0.2} \times 1000$$

$$\Lambda_m = 8\,S\,cm^2\,mol^{-1}$$

(73) Answer : (3)

Solution:

Atom	Electronegativity
Si	1.8
C	2.5

and electron gain enthalpy of F = - 328 kJ/mole and for Br = - 325 kJ/mole

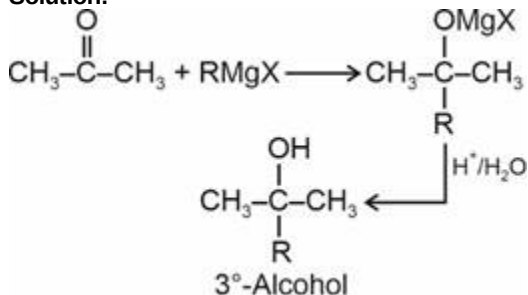
(74) Answer : (1)

Solution:

Since molecularity is number of molecules participating in reaction hence it cannot be zero or fractional.

(75) Answer : (1)

Solution:



(76) Answer : (2)

Solution:

Sulphide ion gives rotten egg like smell as H_2S gas is evolved.

(77) Answer : (4)

Solution:

Group-I – dilute HCl

Group-II – H_2S + dilute HCl

Group-III – NH_4OH + NH_4Cl

Group-IV – $\text{H}_2\text{S} + \text{NH}_4\text{OH}$

(78) Answer : (4)

Solution:

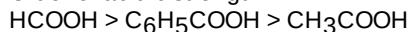
$$\begin{aligned} t_{93.75\%} &= 4 \times t_{50\%} \\ &= 4 \times 24 \\ &= 96 \text{ min} \end{aligned}$$

(79) Answer : (3)

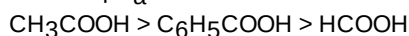
Solution:

As acidic strength increases value of pK_a decreases.

Order of acidic strength



Order of pK_a



(80) Answer : (1)

Solution:

$$\Delta_a H^\circ (\text{Vanadium}) = 515 \text{ kJ mol}^{-1}$$

Copper in +1 oxidation state shows disproportionation in aqueous solution

$$\text{Mn}^{2+}; \mu_{\text{obs}} = 5.96 \text{ BM}$$

(81) Answer : (1)

Solution:

Since density of water is 1 g/mL.

$\therefore$ 18 mL of water weighs 18 g

$$\therefore \text{Moles} = \frac{18 \text{ g}}{18 \text{ g mol}^{-1}} = 1 \text{ mol}$$

$\therefore$ Molecules of water = N_A molecules

(82) Answer : (2)

Solution:

Molar heat capacity is an intensive property.

$\therefore$ For ideal gas $C_p - C_v = R$; irrespective of the moles of the ideal gas.

(83) Answer : (2)

Solution:

• Zr(160 pm) and Hf (159 pm) have almost identical radii

• $\text{Ce}^{4+} (f^0)$ and $\text{Yb}^{2+} (f^{14})$ have no unpaired electrons so they are diamagnetic in nature.

(84) Answer : (3)

Solution:

During adiabatic expansion of an ideal gas, cooling takes place.

$$\therefore T_C < T_A$$

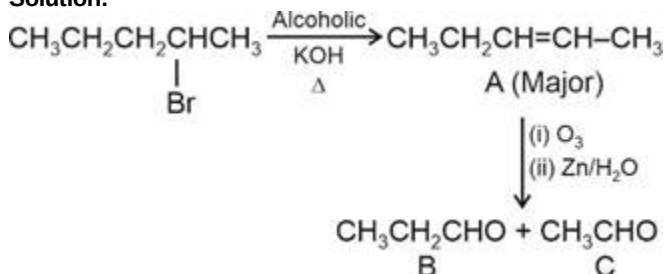
And $w_{\text{isothermal}} > w_{\text{adiabatic}}$

During isothermal process $\Delta T = 0$

$$\therefore T_A = T_B$$

(85) Answer : (2)

Solution:



(86) Answer : (2)

Solution:

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{5 \times 10^{-6} \times 2000} = 6.6 \times 10^{-32} \text{ m}$$

(87) Answer : (4)

Solution:

Vitamins Deficiency diseases

Pyridoxine – Convulsions

Thiamine – Beri beri

Riboflavin – Cheilosis

Ascorbic acid – Scurvy

(88) Answer : (1)

Solution:

Amylopectin is branched chain polymer of α -D-glucose units in which chain is formed by C1 – C4 glycosidic linkage whereas branching occurs by C1 – C6 glycosidic linkage.

(89) Answer : (2)

Solution:

Cyclic, planar species having $(4n + 2)\pi$ electrons in conjugation are aromatic in nature.

(90) Answer : (4)

Solution:

For multielectronic species, higher the value of $(n + 1)$, higher is the energy of orbital. For same value of $(n + 1)$, higher the value of n , higher is the energy of the orbital.

BIOLOGY

(91) Answer : (1)

Solution:

Musca domestica – Diptera

Mangifera indica – Sapindales

Solanum melongena – Polymoniales

Triticum aestivum – Poales

(92) Answer : (1)

Solution:

R.H. Whittaker proposed five kingdom classification.

(93) Answer : (3)

Solution:

The Mycoplasma are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen.

(94) Answer : (3)

Solution:

Zaminkand is an example of modified stem which is called corm and it grows vertically beneath soil surface.

(95) Answer : (3)

Solution:

Grapes are the example of parthenocarpic fruit.

(96) Answer : (2)

Solution:

Kingdom Monera includes the most ancient, the smallest, the simplest and the most abundant micro-organisms. It includes prokaryotic organisms. They have 70S ribosomes (50S + 30S subunits). Cyanobacteria are photosynthetic autotrophs.

(97) Answer : (3)

Solution:

In racemose inflorescence, the shoot axis continues to grow indefinitely and the flowers are borne in an acropetal succession i.e., younger flowers are present towards the apex and older flowers are present at the base. In cymose inflorescence, the flowers are borne in a basipetal order.

(98) Answer : (4)

Solution:

'X' represents the epicotyl hook.

(99) Answer : (4)

Solution:

Thinning of cotton is facilitated by ethylene. Cotton shows plasticity.

(100) Answer : (4)

Solution:

Chalaza represents the basal part of the ovule

(101) Answer : (3)

Solution:

A-Tapetum, B-Microspore mother cell, C-Endothecium

(102) Answer : (3)

Solution:

Distribution of bryophytes and pteridophytes is limited because of the need of water for transport of male gametes and fertilization. They lack pollination

(103) Answer : (4)

Solution:

Constricted pod shape is a recessive trait of the pea plant according to the Mendelian studies.

(104) Answer : (3)

Solution:

The genotypic ratio in the F_2 generation for a monohybrid cross is 1 : 2 : 1.

(105) Answer : (4)

Solution:

The trait under study is autosomal recessive in nature. Thus, all the affected individuals are homozygous recessive in nature. I-5 (homozygous recessive) marries with normal homozygous male, all the progenies will be unaffected.

(106) Answer : (1)

Solution:

A closer examination reveals that the relative abundance of carbon and hydrogen with respect to other elements is higher in any living organism than in Earth's crust.

Element	% Weight of	
	Earth's crust	Human body
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	Very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	Negligible

(107) Answer : (4)

Solution:

Inorganic compounds like sulphate, phosphate, etc., are seen in the acid-soluble fraction. The acid-insoluble fraction has only four types of organic compounds i.e., proteins, nucleic acids, polysaccharides and lipids.

(108) Answer : (4)

Solution:

Every 100 mL of oxygenated blood delivers about 5 mL O_2 to tissues under normal physiological conditions but can deliver up to 15 mL O_2 to tissues during strenuous exercise.

(109) Answer : (4)

Solution:

The squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs and are involved in functions like forming a diffusion boundary.

(110) Answer : (1)

Solution:

A nucleotide has three chemically distinct components. One is a heterocyclic compound, the second is a pentose sugar and the third, a phosphoric acid.

(111) Answer : (2)

Solution:

Animals belonging to the phylum Chordata are fundamentally characterised by the presence of a notochord, a dorsal hollow nerve cord and paired pharyngeal gill slits. They have ventral heart and post-anal part (tail).

(112) Answer : (3)

Solution:

Animals accumulate ammonia, urea, uric acid, CO_2 , water and ions like Na^+ , K^+ , Cl^- , phosphate, sulphate, etc., either by metabolic activities or by other means like excess ingestion. Ammonia, as it is readily soluble, is generally excreted by diffusion across body surfaces or through gill surfaces (in fish) as ammonium ions. Kidneys do not play any significant role in its removal.

(113) Answer : (2)

Solution:

Dinosaurs were dominant during Jurassic period of Mesozoic era.

(114) Answer : (2)

Solution:

In 1953, S.L. Miller created electric discharge in a closed flask containing methane (CH_4), hydrogen (H_2), ammonia (NH_3) and water vapour (H_2O) at 800°C .

(115) Answer : (1)

Solution:

The first non-cellular forms of life could have originated 3 billion years back. They would have been giant molecules (RNA, proteins, polysaccharides, etc.). Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life.

The first cellular form of life did not possibly originate till about 2000 million years ago.

(116) Answer : (3)

Solution:

Deletions and insertions of base pairs of DNA cause frame-shift mutations.

(117) Answer : (3)

Solution:

Phloem parenchyma is absent in most of the monocotyledons.

(118) Answer : (3)

Solution:

Hypodermis is absent in roots. In roots, pericycle is made up of parenchymatous cells.

(119) Answer : (3)

Solution:

5 S rRNA is transcribed by RNA polymerase III.

(120) Answer : (1)

Solution:

The repressor protein binds to the operator region of the operon and prevents RNA polymerase from transcribing the operon. DNA fingerprinting works on principle of polymorphism in DNA sequences.

(121) Answer : (2)

Solution:

Nuclein was first identified by Friedrich Meischer in 1869. RNA mutates at a faster rate. *Caenorhabditis elegans* is a free-living non-pathogenic nematode whose DNA is already sequenced.

(122) Answer : (3)

Solution:

Lambda phage has dsDNA with 48502 bp. UTRs are present at 5'-end before start codon and 3'-end after stop codon. Maurice Wilkins and Rosalind Franklin had not proposed a definitive model for DNA.

(123) Answer : (4)

Solution:

All tissues except epidermis and vascular bundles constitute the ground tissue.

(124) Answer : (3)

Solution:

There are no tRNAs for stop codons.

(125) Answer : (2)**Solution:**

Theodore Schwann reported that animal cells have a thin outer layer known as the plasma membrane.

(126) Answer : (3)**Solution:**

Polysomes or polyribosomes consists of several ribosomes attach to a single mRNA and form a chain. The ribosomes of polyribosome translate the mRNA into protein.

(127) Answer : (3)**Solution:**

- (i) Integral proteins are partially or totally buried in the membrane.
- (ii) Integral membrane proteins require crude method of treatment like detergents.

(128) Answer : (2)**Solution:**

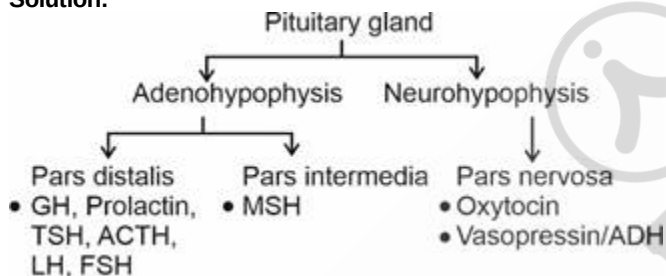
Cell repair is a significance of mitosis.

(129) Answer : (3)**Solution:**

Most of the organelle duplicates in post-mitotic phase whereas centrosome, which is an organelle duplicates during the S-phase and interkinesis. Mitochondria, chloroplast and Golgi bodies duplicates in pre-mitotic (G_2) phase.

(130) Answer : (1)**Solution:**

During the S-phase, chromosome number remains the same, but DNA content doubles.

(131) Answer : (3)**Solution:****(132) Answer : (2)****Solution:**

- Ascariasis – Blockage of intestinal passage, internal bleeding.
- Filariasis – Blockage of lymphatic vessels.
- Amoebiasis – Stools with excess mucous and blood clots.

(133) Answer : (3)**Solution:**

Humoral immune response and cell mediated immune response are types of acquired immunity in which former is antibody mediated while latter is T-lymphocyte mediated.

(134) Answer : (3)**Solution:**

- AIDS is caused due to HIV while cancer can be caused by physical, chemical or biological agents.
- Neither AIDS nor cancer spreads by mere touch or physical contact. AIDS spreads via body fluids and cancer is a non-communicable disease.
- Radiotherapy and immunotherapy are only used for treatment of cancer while AIDS has no cure, only prevention is the best option for it.

(135) Answer : (2)**Solution:**

Morphine is extracted from the latex of poppy plant, *Papaver somniferum*. It is a very effective sedative and painkiller and is very useful in patients who have undergone surgery.

(136) Answer : (4)**Solution:**

The testes are situated outside the abdominal cavity within a pouch called scrotum. The male reproductive system is located in pelvis region.

The function of the scrotum is to protect the testes and to keep them at a temperature few degrees below the normal body temperature.

(137) Answer : (3)

Solution:

Fusion of male and female gamete is called fertilization. Embryonic development is called gestation.

(138) Answer : (3)

Solution:

Androgens are produced by Leydig cells under the effect of LH. Leydig cells are situated outside the seminiferous tubules. Formation of zygote is followed by the formation and development of blastocyst.

(139) Answer : (4)

Solution:

A plasmid is autonomously replicating, circular and extra-chromosomal DNA. Some plasmids carry antibiotic-resistant genes.

(140) Answer : (3)

Solution:

Bioreactors are large vessels in which raw materials are biologically converted into specific products, individual enzymes, etc., using microbial plant, animal or human cells. Sampling port, flat-blade impeller, oxygen delivery system, etc., will be present in both types of bioreactors; but sparger is exclusively present in sparged stirred-tank bioreactor. In a sparged stirred-tank bioreactor, sterile air bubbles are sparged through the sparger.

(141) Answer : (3)

Solution:

The three main steps of PCR are

- Denaturation
- Annealing
- Extension

If the extension step of the PCR gets hindered, *Taq* polymerase will not act on the primers, hence, there will be no extension. Thus, the desired DNA fragment will not get amplified.

Primers attach to the 3' end of the template strand during the annealing step of PCR.

Single-stranded DNA templates are generated by the denaturation of the source DNA during the denaturation step of PCR.

(142) Answer : (3)

Solution:

EcoRI cuts the DNA between bases G and A only when the sequence GAATTC is present in the DNA.

(143) Answer : (2)

Solution:

The human brain is well protected by the skull. Inside the skull, the brain is covered by cranial meninges consisting of an outer layer called dura mater, a very thin middle layer called arachnoid and an inner layer (which is in contact with the brain tissue and spinal cord) called pia mater.

(144) Answer : (3)

Solution:

Myasthenia gravis is an autoimmune disorder affecting neuromuscular junction leading to fatigue, weakening, and paralysis of skeletal muscles. After menopause, estrogen level decreases which causes decreased bone mass and increased chances of fractures. Gout is a type of arthritis.

(145) Answer : (4)

Solution:

Atherosclerosis, affects the vessels that supply blood to the heart muscles. In cardiac arrest, heart stops beating. Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

(146) Answer : (3)

Solution:

In a fully contracted skeletal muscle, length of

- H-zone – disappears
- A-band – remains same
- I-band – reduces
- Sarcomere – reduces

(147) Answer : (3)

Solution:

Tetany is characterised by rapid spasms (wild contraction) in muscles due to low Ca^{++} in body fluid.

Parathyroid hormone (PTH) increases Ca^{+2} levels in blood. PTH is a hypercalcemic hormone.

(148) Answer : (1)

Solution:

A patch of nodal tissue is present in the right upper corner of the right atrium called sino-atrial node (SAN). It can generate the maximum number of action potentials, *i.e.*, $70 - 75 \text{ min}^{-1}$. It is also called the pacemaker of human heart.

(149) Answer : (3)

Solution:

Fibrins are formed by the conversion of inactive fibrinogen in the plasma by the enzyme thrombin. Thrombin, in turn are formed from another inactive substance present in the plasma called prothrombin. An enzyme complex, thrombokinase, is required for the above reaction.

(150) Answer : (1)

Solution:

The smooth muscle fibres taper at both ends (fusiform) and do not show striations. Cell junctions hold them together and they are bundled together in a connective tissue sheath. In cardiac muscles, communication junctions (intercalated discs) at some fusion points allow the cells to contract as a unit, *i.e.*, when one cell receives a signal to contract, its neighbours are also stimulated to contract.

(151) Answer : (4)

Solution:

Cellulose does not contain complex helices and hence, cannot hold I_2 .

(152) Answer : (4)

Solution:

Gastrin – Acts on the gastric glands and stimulates secretions of hydrochloric acid and pepsinogen.

Secretin – Acts on exocrine pancreas and stimulates secretion of H_2O and bicarbonate ions.

CCK – Acts on both pancreas and gall bladder and stimulates the secretion of pancreatic enzymes and bile juice respectively.

GIP – Inhibits gastric secretion and motility.

(153) Answer : (4)

Solution:

Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes like urea and uric acid for water conservation. Marine fishes are ureotelic animals. Reptiles, birds, land snails and insects are uricotelic animals. They excrete nitrogenous wastes in the form of pellet or paste with a minimum loss of water.

(154) Answer : (2)

Solution:

Earth was supposed to have been formed about 4.5 billion years back (4500 million years ago).

(155) Answer : (2)

Solution:

Release of sperms into the seminiferous tubules from Sertoli cells is called spermiation. Transformation of spermatids into spermatozoa is called spermiogenesis. Changes in sperms within the female genital tract before fertilisation is known as capacitation and release of semen by penis into vagina is called insemination.

(156) Answer : (3)

Solution:

T.W. Engelmann worked on green alga *Cladophora* with aerobic bacteria and based on his findings, first action spectrum of the photosynthesis was described.

(157) Answer : (1)

Solution:

Chlorophyll *b* shows yellow-green colour, chlorophyll *a* shows bright or blue-green, carotenoids show yellow to yellow - orange colour in the chromatogram.

(158) Answer : (1)

Solution:

Decarboxylation occurs only during alcoholic fermentation. No CO_2 is released during lactic acid fermentation.

(159) Answer : (4)

Solution:

Plants → Grasshoppers → Frog → Snake
20,000 J 2000 J 200 J 20 J

(160) Answer : (3)

Solution:

Succinate dehydrogenase enzyme is common in both Krebs' cycle and mitochondrial electron-transport system.

(161) Answer : (2)

Solution:

The expression to evaluate the net primary productivity is $GPP - R = NPP$

(162) Answer : (3)

Solution:

Cycas possess coralloid roots.

(163) Answer : (3)

Solution:

In the members of chlorophyceae, chlorophyll *a* and *b* are the major pigments. They store food in the form of starch and the cell wall is composed of cellulose. *Chlamydomonas* belongs to this group.

(164) Answer : (2)

Solution:

Predominant stage of the life cycle of moss is gametophyte. The sporophyte consists of foot, seta and capsule.

(165) Answer : (4)

Solution:

Microbe used in dosa and idli preparations – *Leuconostoc* and *Streptococcus* species of bacteria.

(166) Answer : (3)

Solution:

Swiss cheese is ripened with the help of bacterium called *Propionibacterium sharmanii*.

(167) Answer : (4)

Solution:

Competition usually occur for common space and food.

(168) Answer : (3)

Solution:

As we move from low to high latitude, biodiversity decreases.

(169) Answer : (2)

Solution:

In an airplane, all parts are joined together using thousands of rivets (species).

(170) Answer : (1)

Solution:

Pacific salmon fish and some bamboo species breed once in their life time.

(171) Answer : (4)

Solution:

Spermathecae are present in female cockroach in the 6th abdominal segment. In a male cockroach, testes lie laterally in the 4th-6th abdominal segments. Anal styles are found attached with the 9th abdominal segment. The 10th segment bears anal cerci in both male and female cockroaches.

(172) Answer : (4)

Solution:

Three major regions make up the brain stem; mid brain, pons and medulla oblongata.

The midbrain is located between the thalamus/ hypothalamus of the forebrain and pons of the hindbrain. A canal called the cerebral aqueduct passes through the midbrain. The dorsal portion of the midbrain consists mainly of four round swellings (lobes) called corpora quadrigemina.

The midbrain receives and integrates visual, tactile and auditory inputs.

(173) Answer : (3)

Solution:

Since, DNA fragments are negatively charged molecules, they can be separated by forcing them to move towards the anode (positive electrode) under an electric field through a medium /matrix.

The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation.

Bromophenol blue is a tracking dye. Wells are made towards the negative electrode (cathode).

(174) Answer : (2)

Solution:

Porifers have porous body. Coelenterates (*Obelia*) have only one opening which acts both as mouth and anus.

(175) Answer : (3)

Solution:

Pleurobrachia – Diploblastic, acoelomate animal

Ancylostoma – Pseudocoelomate

Pila – Presence of true coelom

Euspongia – Exhibits internal fertilisation

(176) Answer : (3)

Solution:

A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.

(177) Answer : (2)

Solution:

In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk (2.4 grams per litre). The milk contained the human alpha- lactalbumin and was nutritionally a more balanced product for human babies than natural cow-milk.

(178) Answer : (3)

Solution:

In method, popularly known as test tube baby programme, ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and are induced to form zygote under simulated conditions in the laboratory. The zygote or early embryos (with up to 8 blastomeres) could then be transferred into the fallopian tube (ZIFT – zygote intra fallopian transfer) and embryos with more than 8 blastomeres, into the uterus (IUT – intra uterine transfer), to complete its further development.

(179) Answer : (3)

Solution:

Except for hepatitis-B, genital herpes and HIV infections, other sexually transmitted diseases are completely curable if detected early and treated properly.

(180) Answer : (4)

Solution:

In the alveoli, where there is high pO_2 , low pCO_2 , lesser H^+ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin, whereas in the tissues, where low pO_2 , high pCO_2 , high H^+ concentration and higher temperature exist, the conditions are favourable for dissociation of oxygen from the oxyhaemoglobin.

Aakash
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